Are free fatty acids effective taste stimuli in humans? Presented at the symposium "The Taste for Fat: New Discoveries on the Role of Fat in Sensor...

❖ ❖ ❖

The popularity of low- and reduced-fat foods has increased as consumers seek to decrease their energy consumption. Fat replacers may be used in fat-reduced products to maintain their sensory properties. However, these ingredients have been largely formulated to replicate nongustatory properties of fats to foods and have only achieved moderate success. There is increasing evidence that fats also activate the taste system and uniquely evoke responses that may influence product acceptance. Work supporting a taste component of fat has prompted questions about whether fat constitutes an additional "primary" or "basic" taste quality. This review briefly summarizes this evidence, focusing on human studies, when possible. Effective stimuli, possible receptors, and physiological changes due to oral fat exposure are discussed. Some studies suggest that there are fatty acid tasters and nontasters and if verified could have implications for targeted product development.